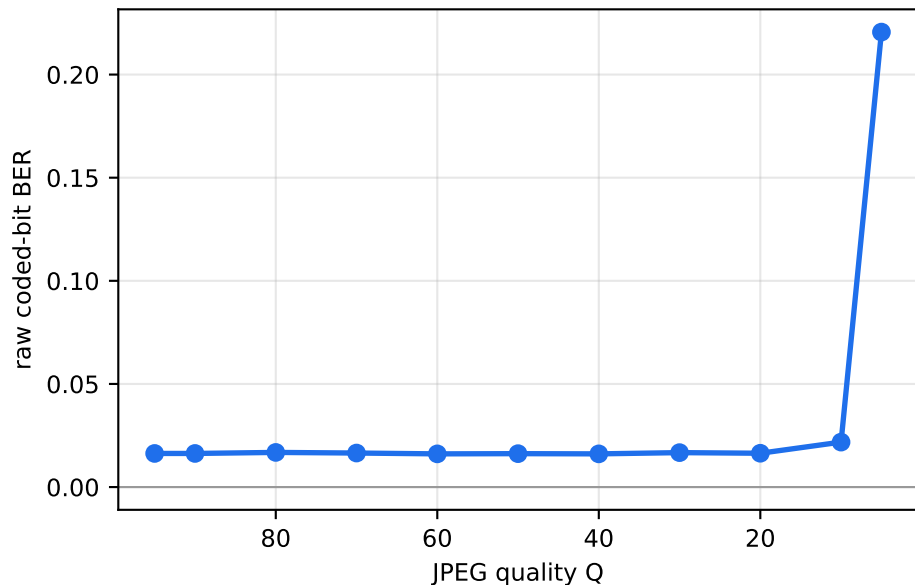
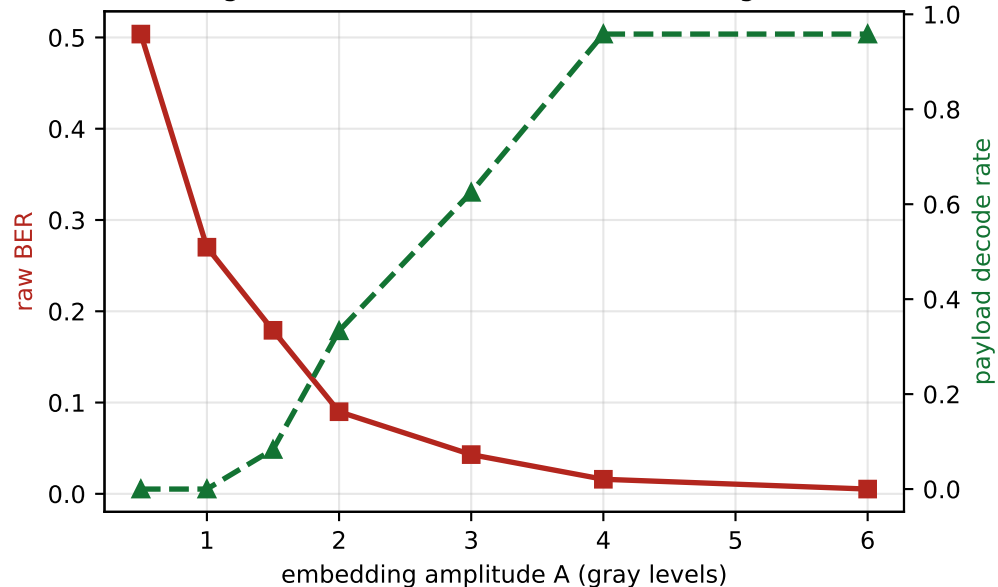


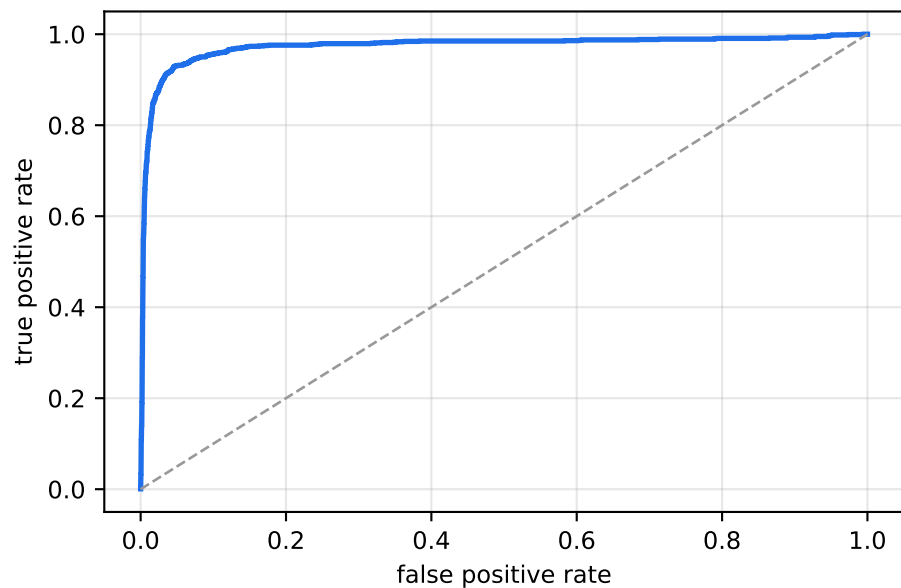
(a) In-pixel channel BER vs real-JPEG quality
(Kodak, pre-ECC; RS recovers payload $\geq Q10$)



(b) Amplitude operating curve @ JPEG Q40
(genuine robustness cliff on real images)



(c) Tamper-localization ROC (real splices)
block-level, AUC ≈ 0.97



(d) Passive PRNU baseline: genuine-impostor
margin shrinks under JPEG (motivates active mark)

